# Personal Statement

## MSc Data Science and Artificial Intelligence for Sustainability

Cranfield University

Working with the real-time data acquisition and aerodynamics department for an electric Formula Student team taught me how valuable data can be for making better decisions. During vehicle testing, monitoring various parameters showed me how quickly engineering decisions can depend on the quality and interpretation of data. I became interested not only in building systems that collect information but also in understanding how analytical methods and artificial intelligence can turn it into reliable and actionable insight. For this reason I am applying for Cranfield University's **MSc in Data Science and Artificial Intelligence for Sustainability** to develop that capability in me and prepare myself for a career in data-driven motorsport engineering.

Being a computer science student, my interest developed through my involvement with my team, E-Formula AshwaRiders, our **SAE India Formula Student electric racing** team, where I served as **Aerodynamics and Data Acquisition Lead**. I contributed to the development of telemetry and data acquisition systems for monitoring vehicle parameters and presenting real-time data on the dashboard. Working with sensors and data gave me practical experience on how I can connect software knowledge with data and telemetry. It also taught me to think critically about data quality, data reliability, system integration, and not just collecting information but also interpreting it correctly. Seeing how data support vehicle development by making precise decisions made me want to explore the analytical methods behind those decisions in greater depth.

My undergraduate degree in **computer science and business systems** at **St. Vincent Pallotti College of Engineering and Technology** provided the computational foundation for my interest in data-driven engineering. Through my studies and professional experience, I developed skills in programming, database management, data processing, and software development. At **PugArch Technology**, I worked with web development and Python-based projects, including building web-scraping pipelines to collect data from multiple websites, store it in CSV and JSON formats, and integrate it with databases and other projects. I also developed websites and dashboards, gaining practical experience in data collection, processing, and presentation. My analytics experience at **Infosys** further strengthened my skills in data analysis, dashboard development using Power BI and Excel, data interpretation, and presenting insights in ways that support informed decision-making. Later, at **HIXAA Technologies**, I worked on LabVIEW-based applications for an audiometer and a Pune Maha Metro simulator, gaining experience in hardware–software integration, real-time product testing, and technical documentation. This role gave me a deeper understanding of how engineering solutions are developed, tested, and delivered to meet practical industry requirements. This experience also helped me to improve the speed of a telemetry and data analysis system using LabVIEW. These experiences taught me to approach technical problems systematically, collaborate with a team, and apply my technical knowledge to practical engineering challenges.

My interest in data-driven problem-solving also led me to **machine-learning research**. Through the **DL-SEC** project on **deep-learning-based** anomaly detection systems using CNN and LSTM approaches, I explored how machine models can identify different patterns that may be difficult to recognize through conventional analysis alone. I learned that developing a model is only one part of meaningful research, but consistent evaluation, experiments, and comparable performance metrics are equally important when assessing whether a model is strong and reliable. This experience strengthened my interest in developing a more rigorous understanding of machine learning, model evaluation, and research methodology.

In motorsport, performance depends on balancing energy availability, energy deployment, optimization of energy, perfect vehicle setup, proper data utilization, and the demands of the vehicle.

Similar challenges arise in wider energy and transport systems, where improving performance must be considered along with use of less resources and environmental impact. I am interested in learning how data science, predictive analytics, and AI can support better energy management, identify problems, and improve decisions in systems where conditions change continuously. **For me, sustainability is not separate from engineering; rather, it offers a broader perspective on how engineering knowledge and skills can be applied to create more efficient, responsible, and environmentally conscious solutions.**

The course **Data Science and Artificial Intelligence for Sustainability** at Cranfield University is interesting to me because it brings together computational and analytical skills with the use of data science and AI to address real-world challenges. My experience in developing telemetry and data-acquisition systems for an electric Formula Student vehicle, alongside my work in software development, data analytics, and developing engineering applications, has shown me how data can help engineers understand system behavior and make better data-driven decisions. Modules such as **Scientific Python**, **Artificial Intelligence for Sustainability**, and **Data Analytics for Sustainability** will help me to build on this foundation by developing stronger skills in analytical modelling, machine learning, and data-driven problem-solving. In particular, I am interested in researching and project work in this course that could give me the opportunity to use these approaches for practical engineering problems related to energy efficiency and sustainability.

I am particularly interested in learning from **Dr. Mennan Guder's** work on machine learning, digital energy systems, and AI for decarbonization. His research and teaching in Artificial Intelligence for Sustainability and Data Analytics for Sustainability align closely with my aim to develop the skills needed to turn complex engineering data into meaningful, actionable insights. My work on electric Formula Student vehicle telemetry helped me gain experience in collecting and analyzing vehicle performance data, which motivated me to explore how predictive modeling and advanced analytics could support more informed engineering decisions. I want to increase my knowledge about machine learning, demand analysis, and optimization techniques and their potential role in improving engineering systems. By combining Cranfield's expertise in AI, energy systems, and sustainability with my practical background in software development and vehicle data acquisition, I aim to build on this knowledge through an applied research project investigating how intelligent telemetry analysis can help engineers make faster, better-informed decisions, with potential applications in vehicle performance and energy management. In the longer term, I aim to contribute to data-driven engineering in motorsport and the wider mobility sector, developing solutions that improve performance while considering energy efficiency and environmental impact.

After the MSc, I aim to pursue a data-focused role in motorsport, ideally within **Formula 1**, in areas such as vehicle performance analysis, telemetry, simulation, or energy systems. I am particularly interested in how data science and AI can help engineers extract reliable insights from real-time vehicle data, investigate performance limitations, and make better-informed engineering decisions. Over the longer term, I want to contribute to engineering approaches that combine competitive performance with more efficient use of energy and resources. The Cranfield MSc program would help me move towards this goal by strengthening my foundations in data science and AI, broadening my understanding of sustainability, and developing my ability to investigate complex engineering problems through rigorous applied research . Ultimately, I wish to combine the worlds of data analysis and practical engineering and contribute towards the motorsport and mobility sector that is not only more competitive but also more intelligent, efficient, and sustainable.